

Spine Formal-v7 Primary Dynamic-Graph Results

Correctness-Gated Spine vs. GraSU+ReGraph K4-shared

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Live snapshot, July 2026

Abstract

This live report contains 25 of 46 expected formal-v7 executions and 9 complete Spine/K4-shared pairs. Missing bars are unfinished, policy-stopped, or behavior-transition-invalidated executions awaiting an identical-case successor, never zero-valued measurements. Every admitted row passes architecture-precision and independent mathematical oracles, full final-state comparison, and request, response, and DRAM conservation.

1 Measurement contract

Algorithm	System	Window	Warm state
CC	G+R K4-shared	complete_execution	False
CC	Spine	complete_execution	False
ResPR	G+R K4-shared	complete_execution	False
ResPR	Spine	complete_execution	False
SSSP	G+R K4-shared	complete_execution	False
SSSP	Spine	dynamic_e2e_to_convergence	True

Table 1: Measurement windows observed in admitted formal-v7 rows.

Dataset scope. The frozen corpus contains 11 real datasets; 9 enter the full-graph comparison. The remaining datasets, BC (24,575,382 vertices), UK (18,520,486 vertices), exceed the frozen Spine capacity of 16,777,216 vertices. They are capacity exclusions, not failed or selectively removed performance rows, and no slices replace them in the full-graph aggregate.

Synthetic endpoint. R19-32 (524,288 vertices and 15,483,485 unique directed edges) is reported as a separate scalability endpoint. It is never included in a real-dataset aggregate; the figures and cycle table place it after a dotted or ruled separator.

SSSP interpretation. Spine starts from a verified persisted old-graph SSSP state and measures the accepted update, automatic active-source discovery, incremental propagation, and drain. The conversion-free GraSU+ReGraph baseline performs its declared complete ReGraph execution because it has no equivalent persisted incremental SSSP state. This is an architecture-level dynamic-service comparison, not an identical-kernel microbenchmark.

Claim boundary. The frozen full matrix remains partial while expected rows are unfinished, policy-stopped, or behavior-transition-invalidated; a row must complete or receive a declared scope exclusion before the matrix can be called complete. Device cycles, accepted memory bytes, and DRAMSim3 HBM energy are simulator outputs. They do not claim cycle-for-cycle FPGA calibration, on-chip dynamic energy, or total board power.

Behavior-transition coverage. The following 1 prior Spine rows are deliberately absent until identical-case successors pass with identical final state: HW-SSSP. They are not failures, zeros, or inputs to any aggregate.

Full-graph wall-time feasibility. An execution-feasibility model fitted to 3 completed, correctness-admitted K4-shared SSSP rows projects total host times of SO 105.7 h, PK 342.2 h at the observed cycle rates. Even a stress test with ten times less work and twice the observed simulator rate leaves SO 3.6 h, PK 16.8 h, compared with the 3 h campaign budget. Those full-graph executions were therefore soft-stopped by policy. Their partial cycles and memory requests are retained only as host-runtime evidence and never enter accelerator-performance aggregates. The validated source-matched preflights report LJ08 source 26, 35 supersteps, 1182.1 h, R19 source 113, 10 supersteps, 52.8 h, LJ source 71, 27 supersteps, 789.2 h. Their total-cycle estimates multiply validated oracle work by the median completed-run cycles/edge-round; they are projections, not completed cycle simulations. For the remaining 2 unlaunched real graphs, a conservative one-round screen uses the smallest completed-run cycles/edge-round and the largest completed-run simulator rate. Even the mandatory first full-graph round projects HW 35.7 h, OK 74.2 h; these screened rows were not launched and are not performance data.

Machine-checked evidence audit. All 25 observed executions pass their individual architecture and mathematical correctness gates, closed AXI/HBM arbitration and traffic ledgers, and DRAM request conservation. Component-level activity is present for all observed executions (163 normalized component rows).

2 Current primary comparison

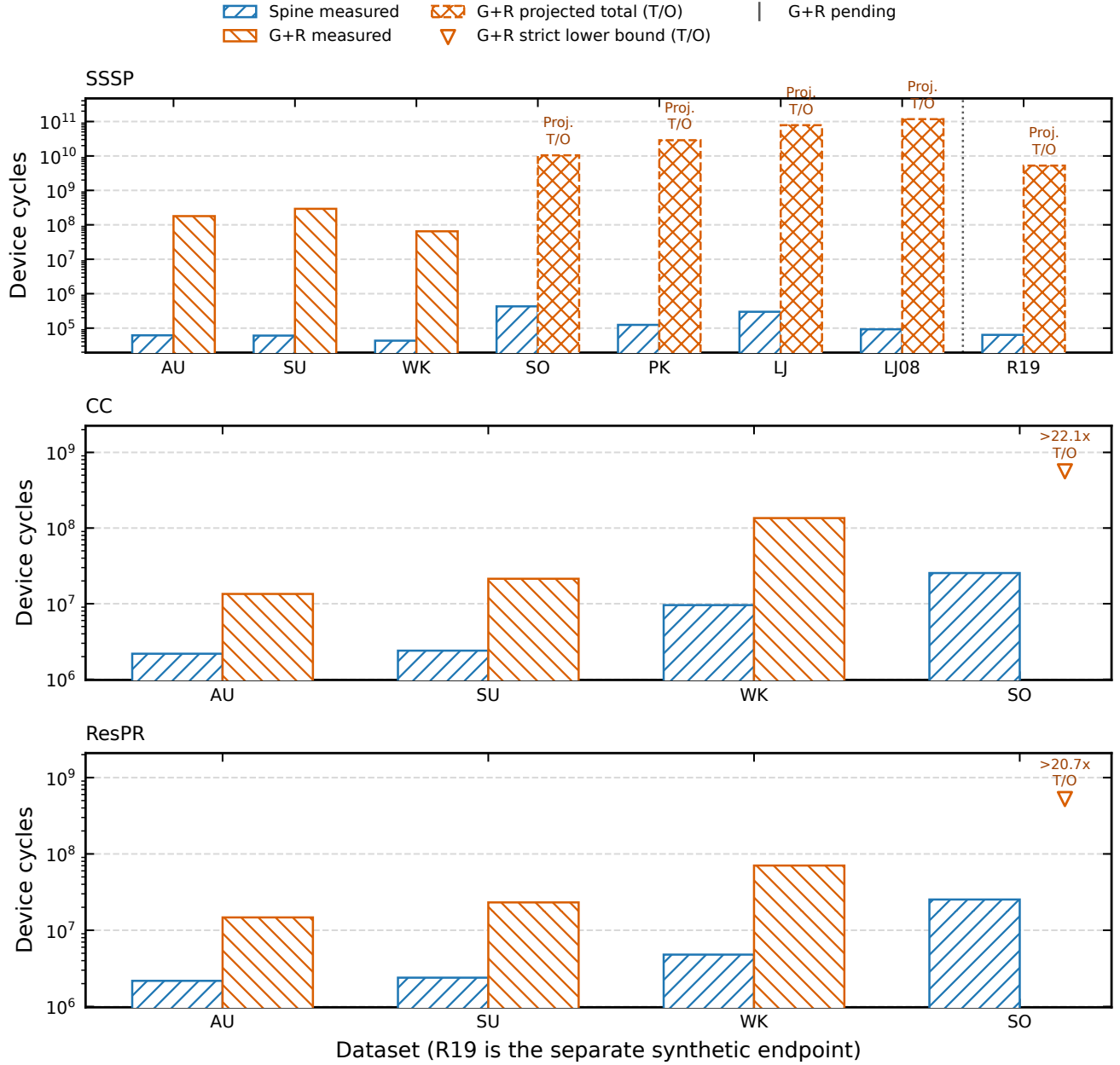


Figure 1: Absolute insertion-batch-8 device cycles for all 16 correctness-admitted current-version Spine rows. Blue bars and unmarked orange bars are completed executions. Cross-hatched G+R bars marked Proj./T/O are total-cycle feasibility projections derived either from stopped execution prefixes or from validated source-matched host preflights; they are not measured performance. Upward triangles are conservative one-round feasibility screens, not total-cycle predictions. Downward triangles for SO CC and ResPR are strict monotonic device-cycle lower bounds from stopped incomplete executions; they are neither completed results nor projected totals. The invalidated prior-version HW row and queued OK row have no admitted Spine result and are therefore absent.

Projection method. For a source-matched preflight row, we compute $C_{\text{proj}} = \text{median}_i \{C_i / (E_i S_i)\} E_{\text{target}} S_{\text{target}}$. The unit cost is fitted from three completed, correctness-admitted K4-shared SSSP executions; the target edge count and minimum superstep count come from the complete-graph host oracle using the same source cohort as Spine. These values are feasibility projections, not completed simulations, and are excluded from measured cross-architecture aggregates.

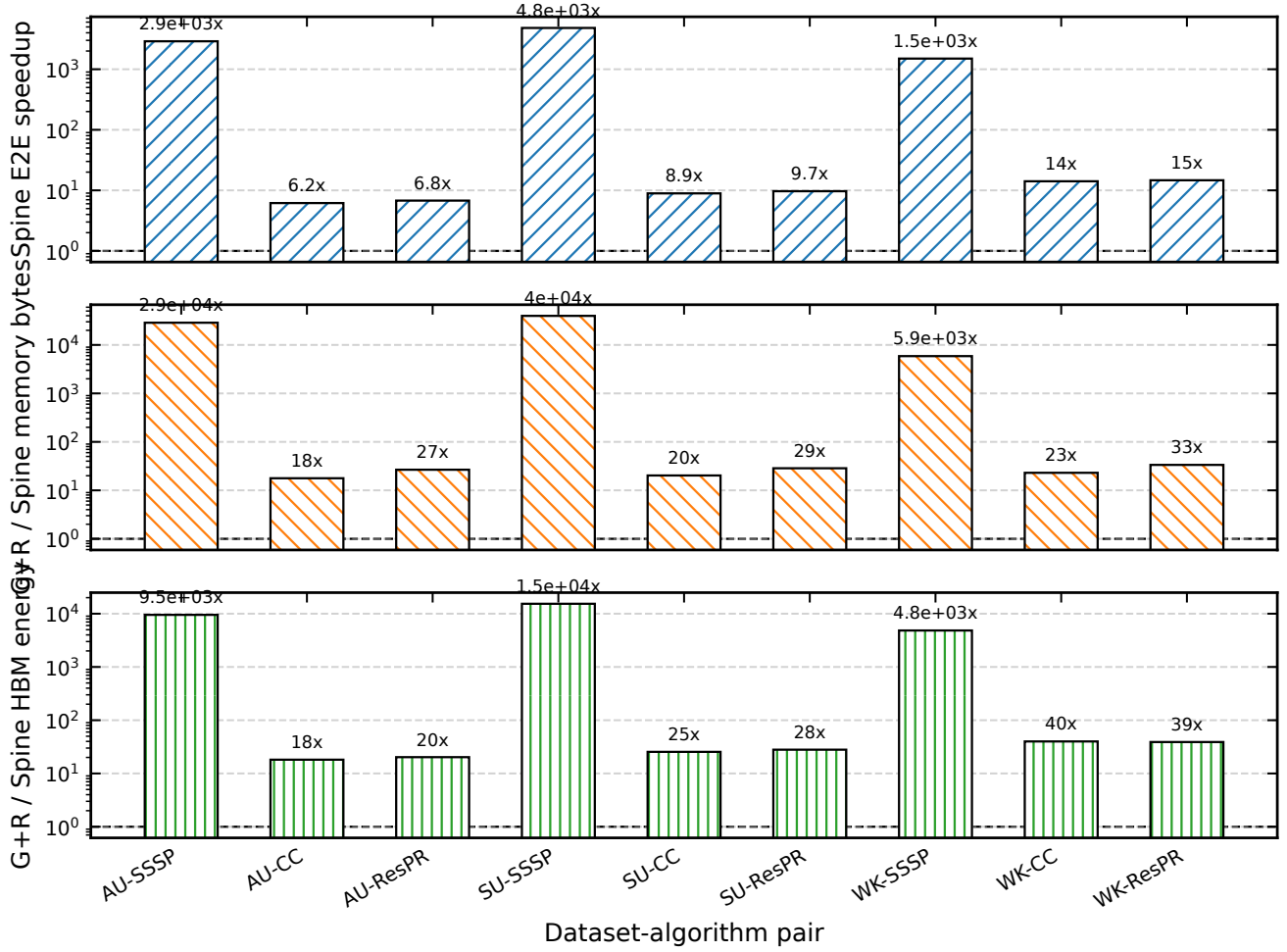


Figure 2: Ratios for complete insertion-batch-8 pairs. Values above one favor Spine for E2E latency and indicate that GraSU+ReGraph uses more memory traffic or HBM energy in the other panels. R19, when complete, appears to the right of the dotted separator and is not part of the real-dataset population.

Data	Algorithm	Spine cyc	G+R K4 cyc	Speedup
AU	SSSP	61,817	179,053,544	2896.51×
AU	CC	2,183,292	13,462,032	6.17×
AU	ResPR	2,175,289	14,729,898	6.77×
SU	SSSP	60,699	291,909,003	4809.12×
SU	CC	2,397,167	21,367,179	8.91×
SU	ResPR	2,392,695	23,162,486	9.68×
WK	SSSP	43,227	64,837,217	1499.92×
WK	CC	9,587,027	135,366,890	14.12×
WK	ResPR	4,796,702	70,417,726	14.68×

Table 2: Absolute device cycles. Only cross-system final-state-matched pairs enter this table. R19 is separated from the real datasets by a rule.

3 Memory traffic and request locality

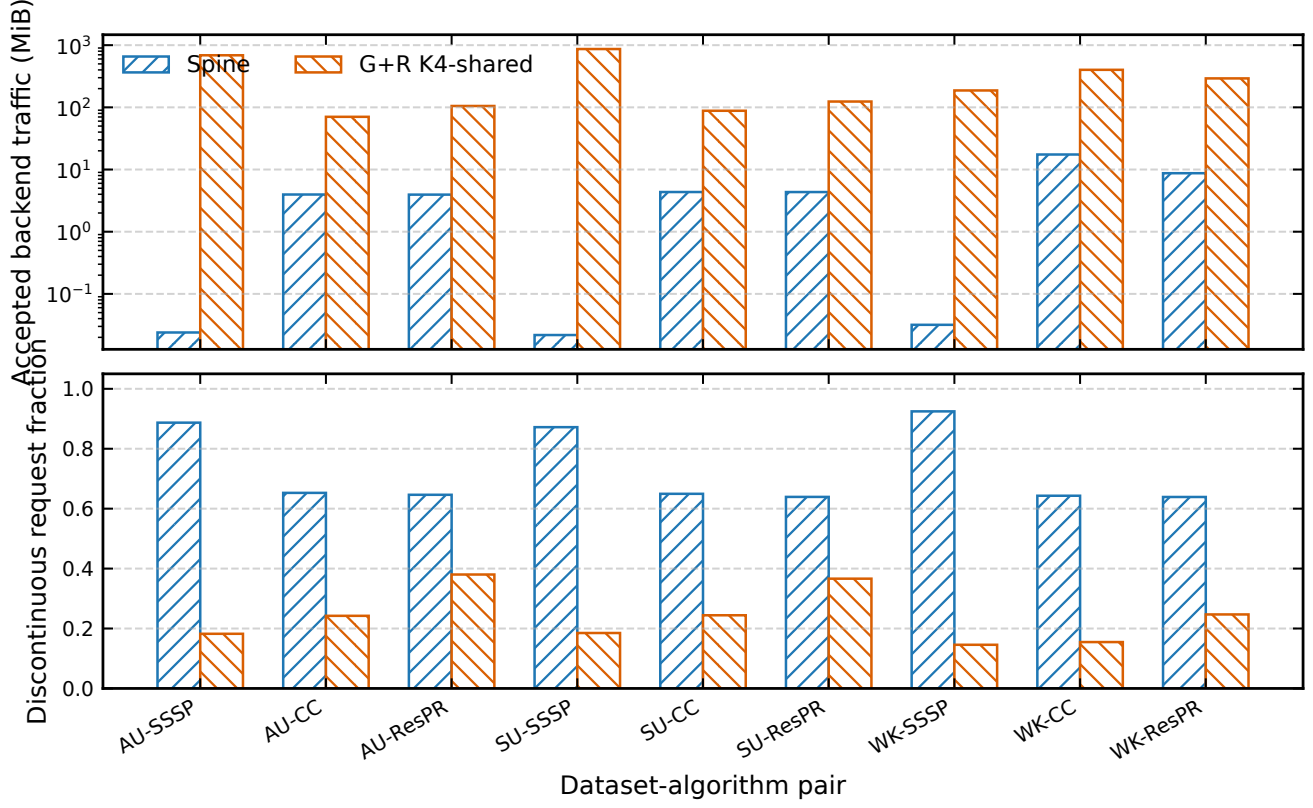


Figure 3: Absolute accepted-backend traffic and request-stream locality for the same complete pairs. A request is discontinuous when its accepted address does not continue the previous request from the same initiator, operation, and logical HBM channel. This is not a DRAM row-buffer-miss metric. R19 is a separate synthetic endpoint.

4 Structure-update throughput

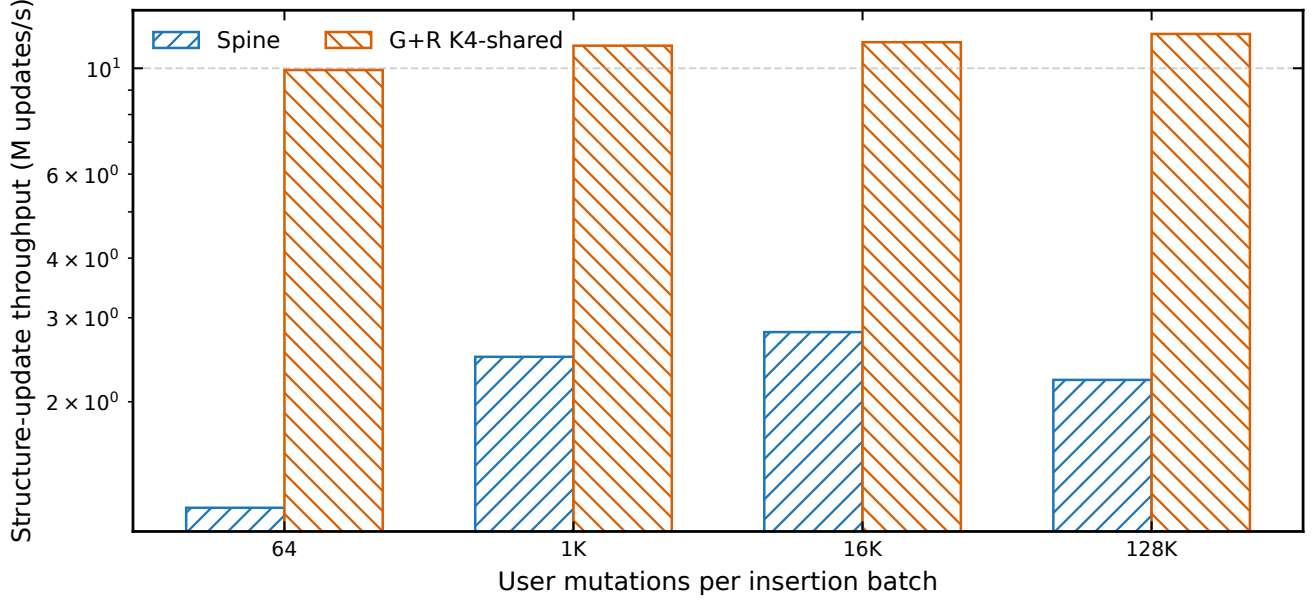


Figure 4: Correctness-gated AskUbuntu full-graph insertion throughput at batch sizes 2^6 , 2^{10} , 2^{14} , and 2^{17} . The current snapshot contains 4 of 4 planned batch points. Spine counts maintenance through its drained completion boundary; G+R counts GraSU PMA update completion. Both exclude graph-algorithm iterations, equivalent to setting max-iter to zero.

Window boundary. This figure isolates graph-structure mutation. The counters come from dedicated correctness-gated executions and stop at each architecture’s serial update-phase boundary. We use source vertex 0, which remains isolated after all four batches, solely to shorten the post-update SSSP correctness drain. Both architectures enforce an update–barrier–compute order, so this source choice cannot affect the recorded update phase. For Spine the phase includes required level selection, L0/carry work, metadata publication, and drain; it excludes reader, propagation, convergence, and algorithm drain. For G+R it includes the GraSU PMA update and excludes ReGraph execution. It must not be read as end-to-end dynamic graph service latency.

Operation coverage. The separate insertion, deletion, and weight-change grid at batches 1, 8, and 64 remains in `update_operation_pairs.csv` and in the fallback/RQ3 evidence. It is not mixed into this scaling plot because non-monotonic SSSP updates exercise a bounded full-rebuild policy rather than direct insertion.

5 Simulator execution cost

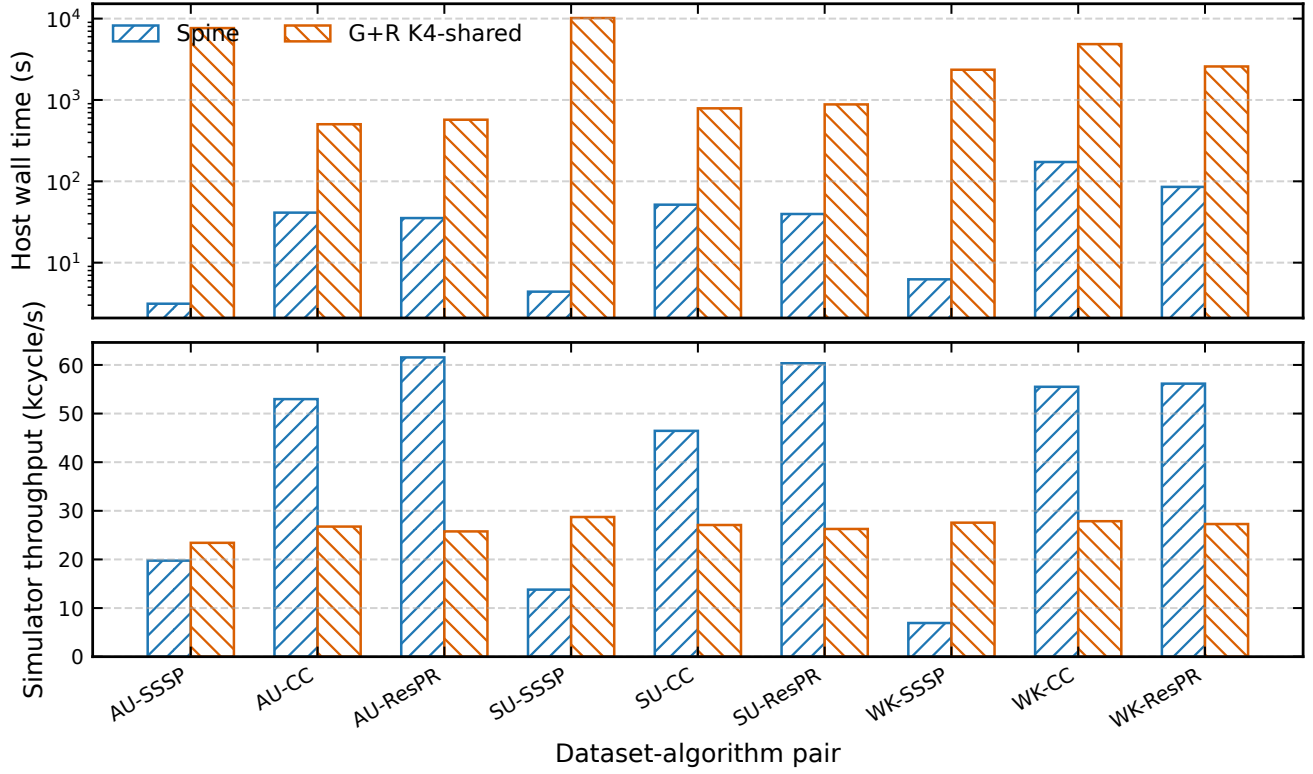


Figure 5: Host wall time and simulated-device-cycle throughput for the correctness-matched pairs. These are simulator engineering diagnostics under the recorded campaign concurrency, not accelerator latency or a cross-architecture performance metric.

Use boundary. Wall time establishes experiment feasibility and identifies simulator optimization targets. Architecture claims use device cycles and the shared memory model; they never use host wall time.

6 RQ3: realized work and latency

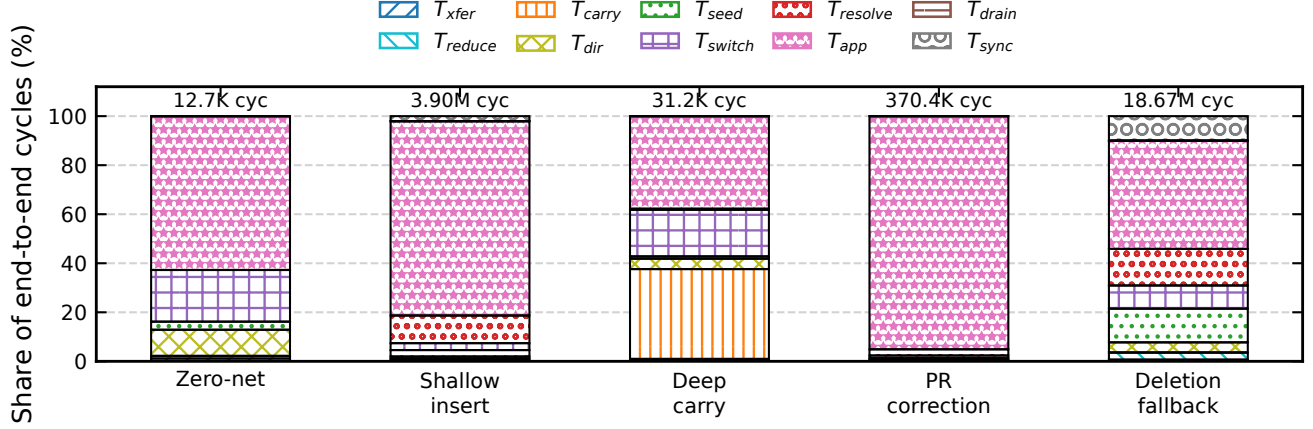


Figure 6: Direct ten-stage, exclusive critical-path composition for zero-net, shallow insertion, deep carry, residual PageRank correction, and SSSP/CC nonmonotonic fallback. The six maintenance stages are counted in the execution core; overlapping resolve/app intervals are assigned to the component gating their completion. For residual PageRank, T_{seed} also contains the serial device correction/seed interval. Every bar closes exactly to E2E device cycles.

Case	Algorithm	E2E cyc	Maint.	Resolve+App	Drain+Sync
Zero-net	CC	12,671	4,720	7,950	1
Shallow insert	SSSP	3,898,885	288,722	3,530,065	80,098
Deep carry	SSSP	31,249	19,316	11,932	1
PR correction	Residual PR	370,417	7,525	361,645	13
Deletion fallback	SSSP	18,668,757	5,773,955	11,024,338	1,870,464

Table 3: Deterministically selected maximum-cycle representative of each eligible realized-work class.

7 RQ3: cost-model validation

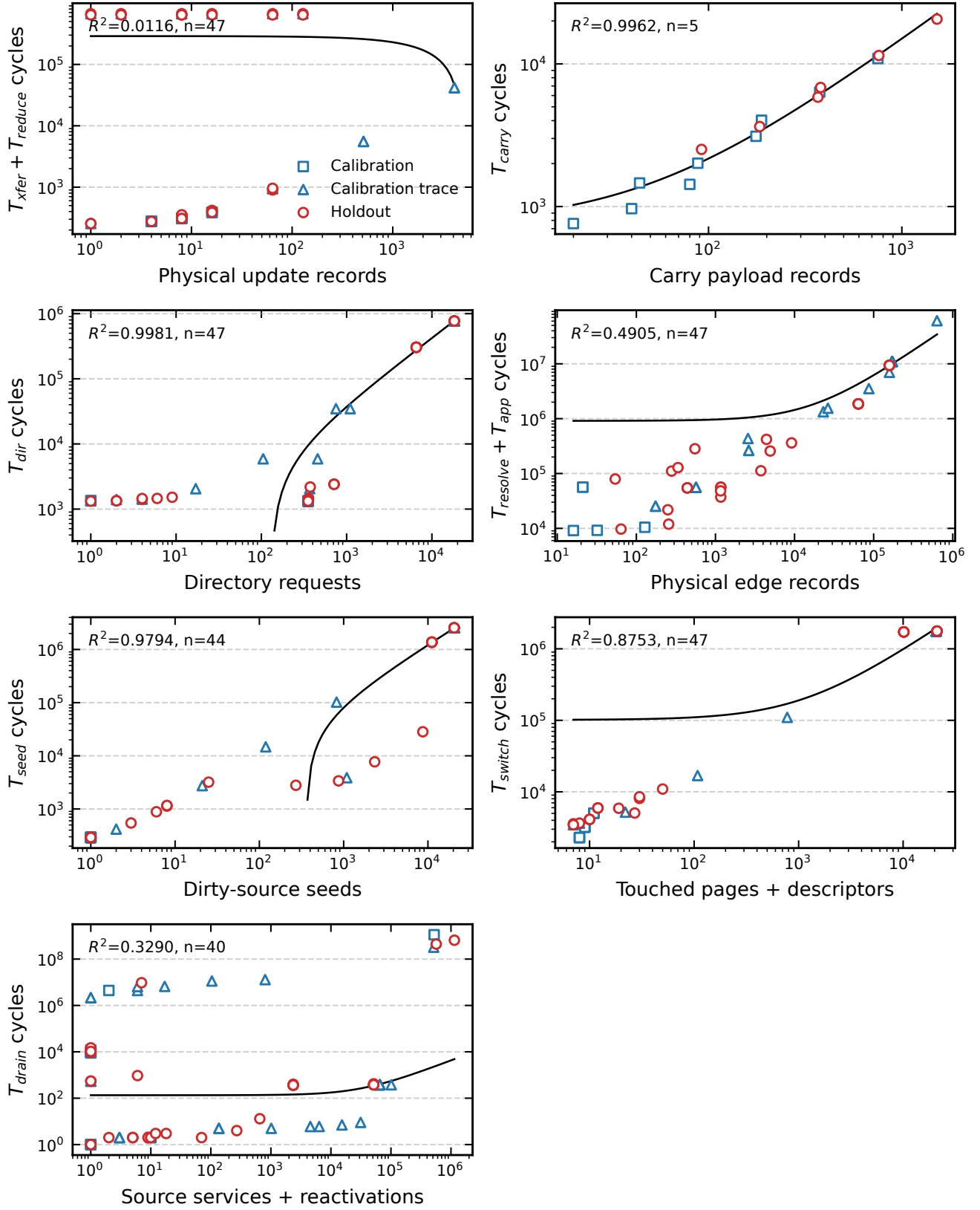


Figure 7: Direct execution counters versus mechanism-local cycles across 95 dual-oracle-admitted executions. Blue squares are calibration cases and red circles are holdout traces.

Mechanism	Samples	Slope	R^2
sort_frontend	47	-59.301	0.0116
carry	5	14.297	0.9962
directory	47	42.736	0.9981
physical_resolve_apply	47	52.950	0.4905
seed	44	125.456	0.9794
switch	47	88.693	0.8753
drain	40	0.004	0.3290

Table 4: Realized-work regressions. Carry work includes old payload reads, merge inputs, and output rewrites; cursor inspection is reported separately.

Interpretation. The seven panels report every requested realized-work relation, including weak sort-only and drain-only relations rather than hiding them. The slope and R^2 table distinguishes mechanisms that transfer across mixed trace classes from those that need a richer topology- or contention-aware predictor.

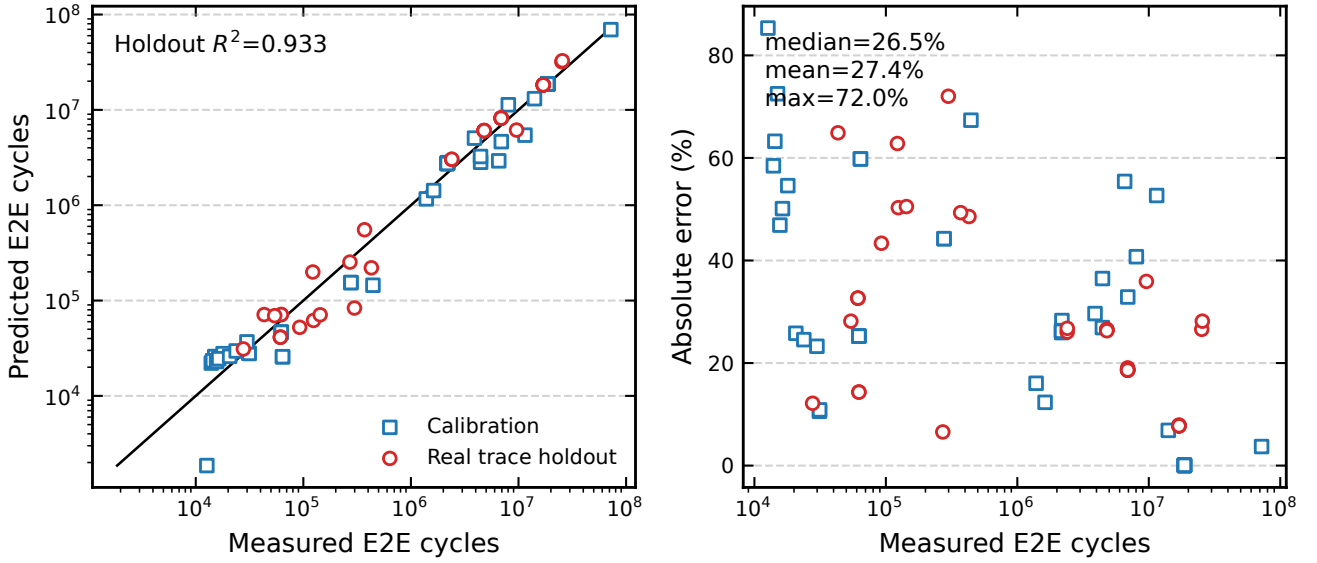


Figure 8: Calibration-only nonnegative realized-work model versus measured E2E cycles, plus absolute residuals. The holdout contains 39 real-trace executions that never participate in fitting; holdout $R^2 = 0.933$, median absolute error 26.5%, mean absolute error 27.4%, and maximum absolute error 72.0%.

RQ3 boundary. These correlations validate internal cost structure and bottleneck attribution; they are not cycle-for-cycle FPGA calibration or proof that one scalar predicts every topology. The timed device boundary starts with an already-resident sorted update buffer: host DMA and an external FLiMS sort are not included. Current T_{seed} measures maintenance dirty-source publication and, for residual PageRank, the serial device correction interval covering resident rank/degree reads, physical-edge scans, seed writes, and active-list publication before propagation. The host supplies only the immutable oracle/work trace.

8 Implementation-level power attribution

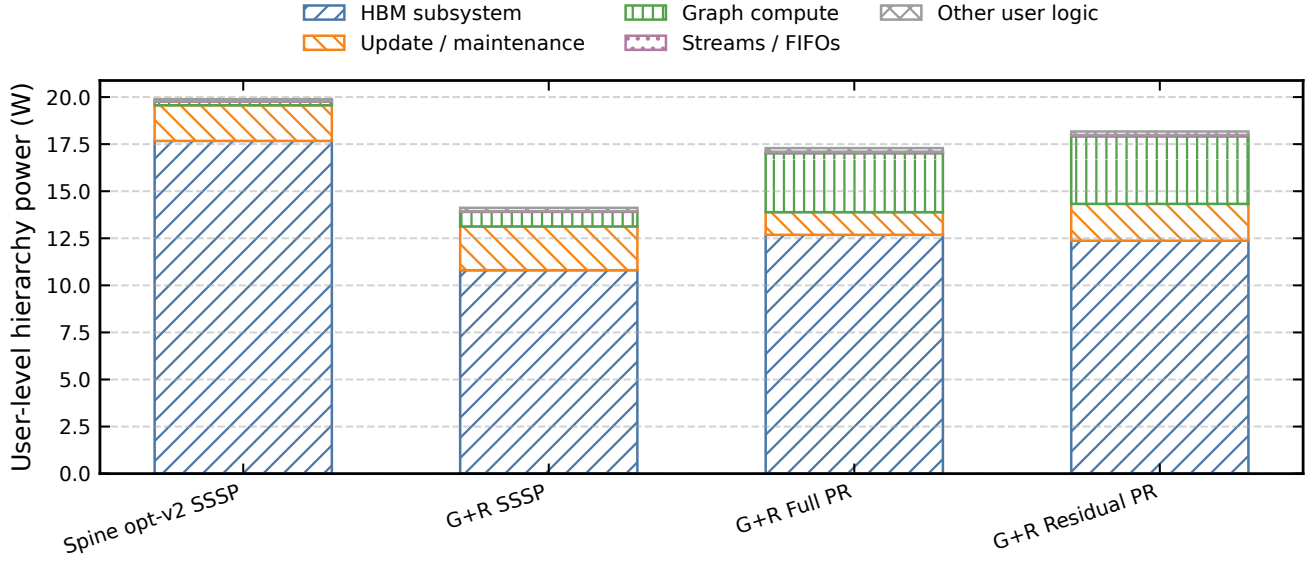


Figure 9: Vivado post-route vectorless hierarchy power at 150 MHz. The four bars are distinct routed builds; the Spine evidence covers SSSP only. Values use Vivado default activity with Low confidence and establish component attribution, not workload-calibrated energy or board power.

Energy boundary. The workload-specific HBM energy ratios in Figure 1 come from DRAMSim3 and are paired with the exact executions plotted there. The routed bars above establish implementation-level component power but retain Vivado’s Low-confidence vectorless activity assumption.

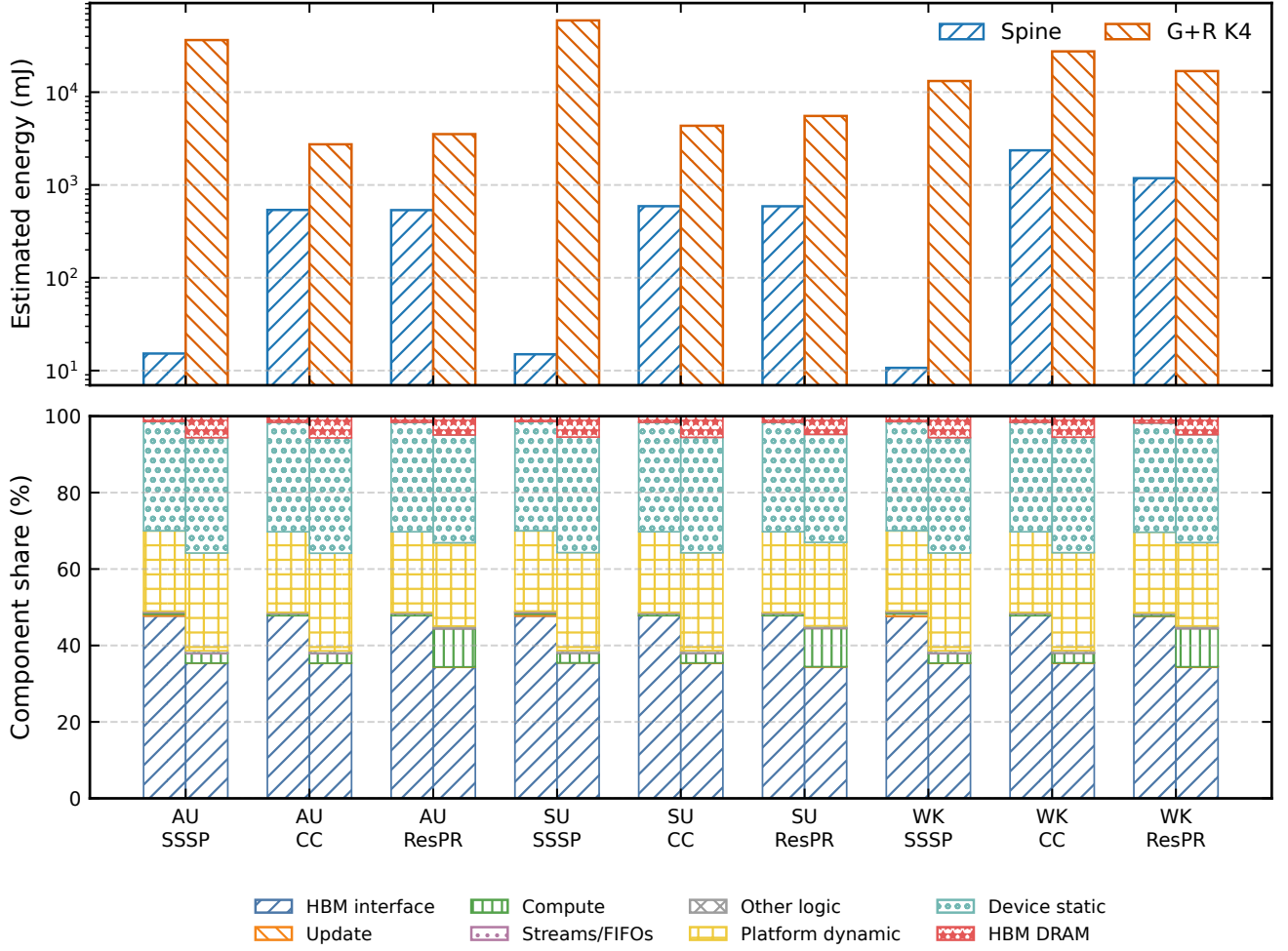


Figure 10: Complete workload-duration energy ledger for the nine matched SSSP, CC, and residual-PageRank pairs. DRAM energy comes from each exact DRAMSim3 command stream. FPGA hierarchy components use execution-driven active cycles multiplied by routed vectorless power; platform dynamic and device static use the full measured interval. All ledgers close. This is a modeled energy estimate, not RTL-SAIF or board-power measurement. Spine CC/Residual and G+R CC use explicitly recorded routed-algorithm proxies.

Energy interpretation. This result has the same separation of concerns used throughout the report: the cycle simulator supplies workload activity and duration, DRAMSim3 supplies HBM-device energy, and implementation tools supply component power. CACTI selected-SRAM ASIC projections remain a separate ledger and are not summed with FPGA BRAM/URAM power. The largest remaining confidence upgrade is algorithm-matched routed builds with RTL-derived SAIF activity.